SBE 9/911plus QA report: SP2613

MPL Internship Program Day at Sea

2026-07-09 to 2026-07-09

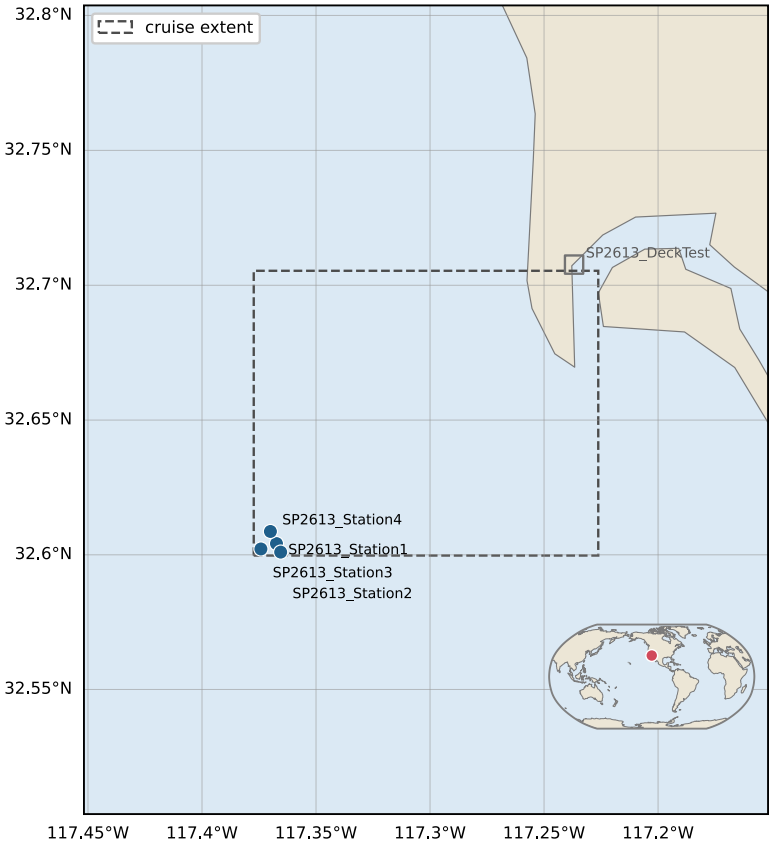
Generated 2026-11-02 09:06 UTC by sbe-qa-processing 0.1.0, seabirdscientific 3.1.1, gsw 3.6.23, numpy 2.5.3

Source: data/SP2613_169847_ctd

All 10 integrity checks pass. 4 of 4 casts warn in the science checks: Sensor calibration age (all 4).

Fileset overview

Cast positions (hollow squares are deck tests; red is outside the extent)



Cast	Scans	Max P [dbar]	Dual T/C	Pass / warn / fail	Notes
SP2613_DeckTest	2588	0	yes	6/0/0	deck test
SP2613_Station1	39865	193	yes	20/1/0	
SP2613_Station2	43853	204	yes	20/1/0	
SP2613_Station3	28730	204	yes	20/1/0	
SP2613_Station4	14187	203	yes	20/1/0	

Pass / warn / fail counts the integrity and science checks.

Integrity checks

Check	Status	Value	Limit	Detail
Checksums match the manifest	pass	20 of 20 files		
Raw files complete	pass	5 of 5 casts pass		
Files readable	pass	5 of 5 casts pass		
Cast position	pass	5 of 5 casts pass		
NMEA position on every scan	pass	5 of 5 casts pass		
Position inside the expected area	pass	4 of 4 casts pass	-117.377 to -117.226 E, 32.5997 to 32.7054 N	
Cast time	pass	5 of 5 casts pass		
Cast time inside the expected dates	pass	4 of 4 casts pass	2026-07-09 to 2026-07-09	
Header serial numbers match the configuration	pass	5 of 5 casts pass		
Casts in time order	pass	file name order is time order	file name order	
Cast times	info	2026-07-09 18:04 to 2026-07-09 20:55 UTC		
Cast positions	info	-117.374 to -117.365 E, 32.601 to 32.6087 N		
Casts	info	5		
Deck-test casts	info	1		SP2613_DeckTest
Casts with bottles fired	info	1		SP2613_Station4
Instrument model	info	SBE 911plus/917plus CTD		

SP2613_Station1

Sensors

SBE 911plus/917plus CTD; 24 Hz; 0 frequency channels and 0 voltage words suppressed; appended: NMEA position

Channel	Sensor	Serial number	Calibrated	Age at cast [days]
Primary temperature (F0)	Temperature	4924	2025-05-08	427
Primary conductivity (F1)	Conductivity	4537	2026-01-06	184
Pressure (F2)	Pressure (Digiquartz)	0381	2025-09-08	304
Secondary temperature (F3)	Temperature	2059	2026-01-30	160
Secondary conductivity (F4)	Conductivity	4605	2026-02-17	142
Voltage 0	Oxygen (SBE 43)	3357	2025-12-13	208
Voltage 1	PAR (Biospherical/Licor/Chelsea)	70444	2024-03-15	846
Voltage 2	Transmissometer (WET Labs C-Star)	CST 0493DR	2026-02-16	143
Voltage 4	Fluorometer (WET Labs ECO-AFL/FL)	FLRTD 9157	2024-10-22	625
Voltage 6	Altimeter	VA500 88426	—	

Ages over 365 days are highlighted.

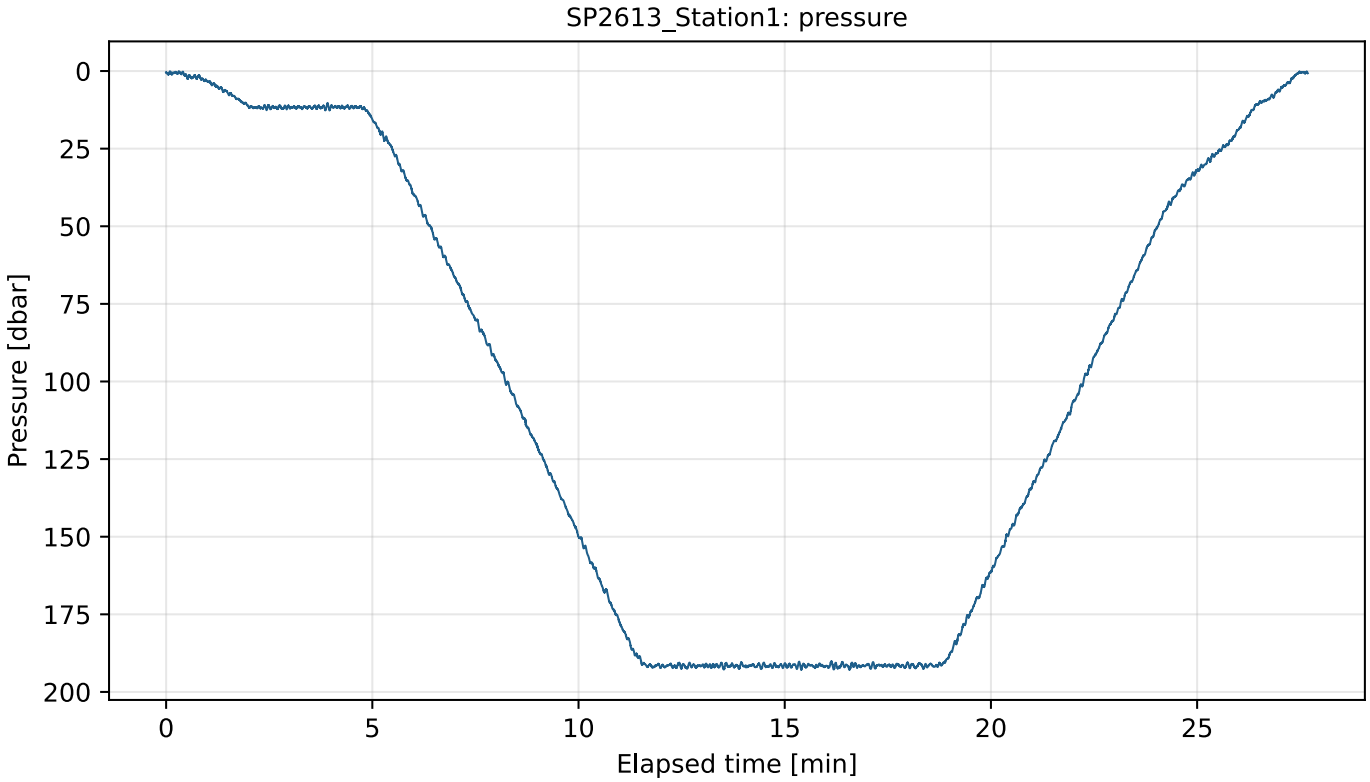
Integrity checks

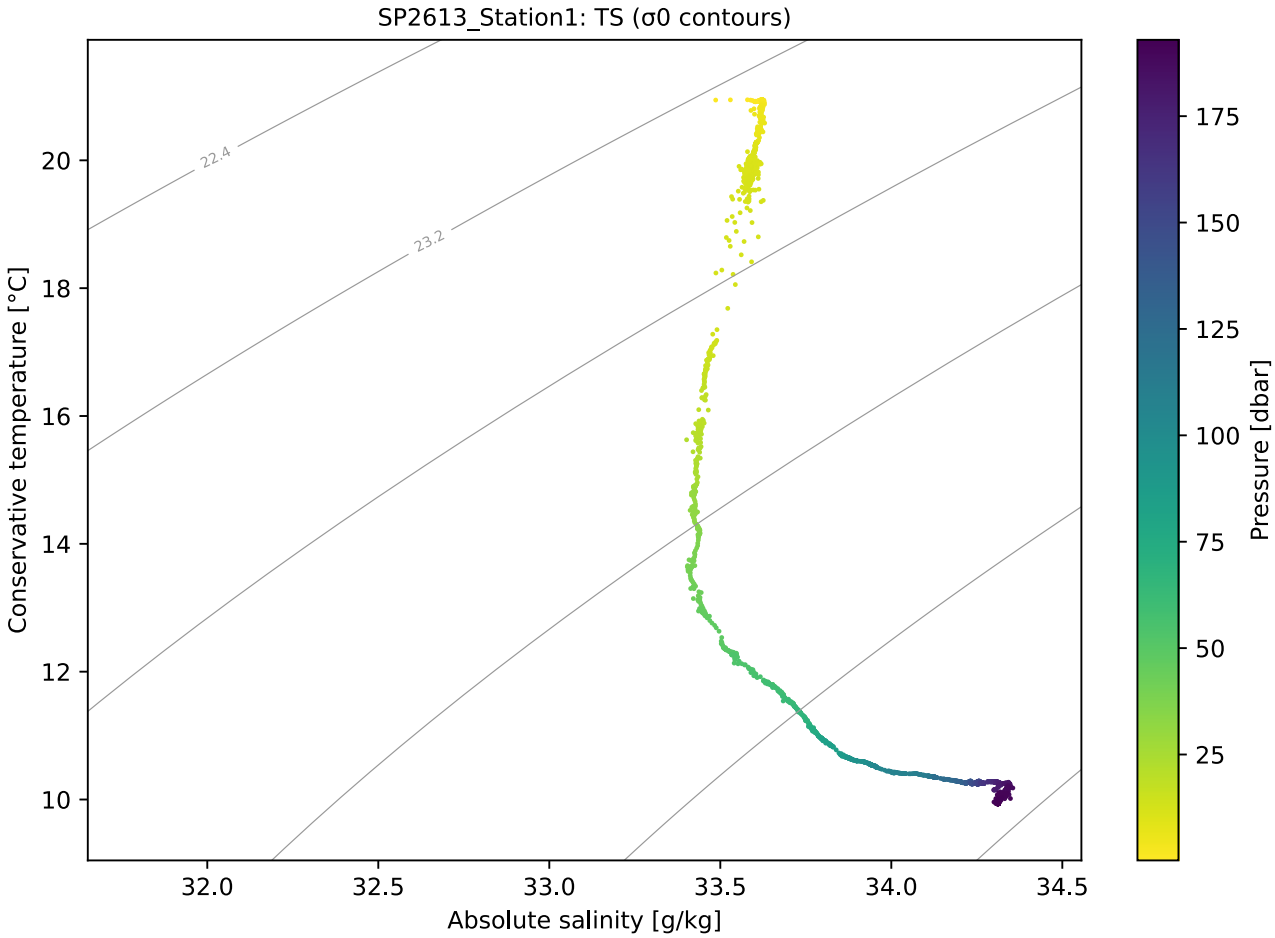
Check	Status	Value	Limit	Detail
Raw files complete	pass	all present		.bl bottle log present
Files readable	pass	39865 scans		
Cast position	pass	32.6042, -117.3673		header
NMEA position on every scan	pass	0 scans without position		
Position inside the expected area	pass	inside	-117.377 to -117.226 E, 32.5997 to 32.7054 N	
Cast time	pass	2026-07-09 18:04:25 UTC		NMEA time
Cast time inside the expected dates	pass	2026-07-09	2026-07-09 to 2026-07-09	
Header serial numbers match the configuration	pass	T 4924, C 4537		

Science checks

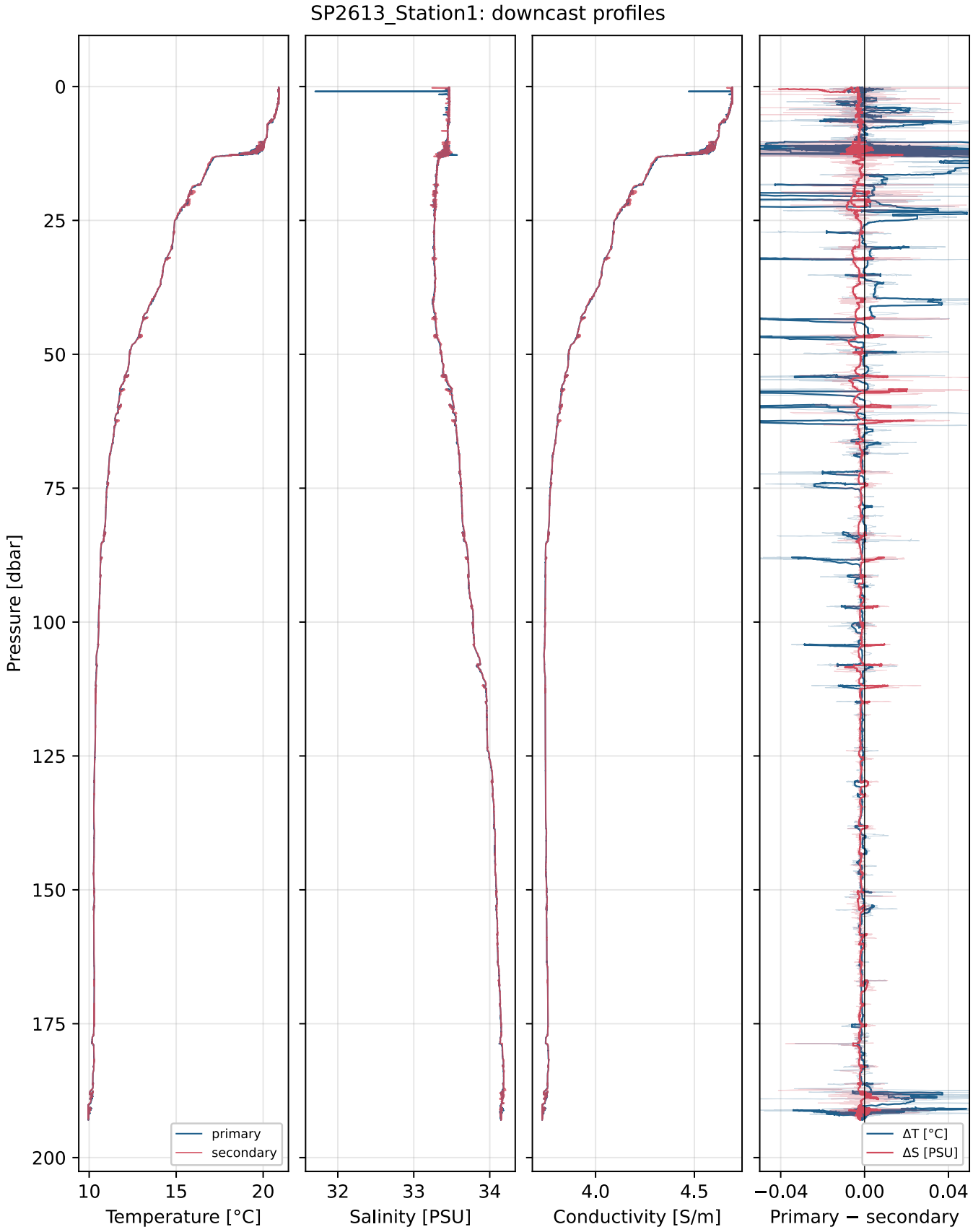
Check	Status	Value	Limit	Detail
Dual temperature difference (median)	pass	0.001826 °C	≤ 0.01 °C	95th percentile 0.04361 °C, below 20 dbar
Dual conductivity difference (median)	pass	0.000348 S/m	≤ 0.001 S/m	95th percentile 0.00371 S/m, below 20 dbar
Dual salinity difference (median)	pass	0.002627 PSU	≤ 0.01 PSU	95th percentile 0.01285 PSU, below 20 dbar
Pressure range	pass	0.00% outside	-5 to 11000	min 0.09608, max 192.9
Temperature range	pass	0.00% outside	-2.5 to 40	min 9.949, max 20.9
Conductivity range	pass	0.00% outside	0 to 7	min 3.728, max 4.692
Salinity range	pass	0.00% outside	2 to 42	min 31.7, max 34.19
Temperature spikes	pass	0.000% of scans	≤ 0.5%	
Conductivity spikes	pass	0.020% of scans	≤ 0.5%	
Pressure spikes	pass	0.000% of scans	≤ 0.5%	
Lost scans (modulo count)	pass	0 (0.000%)	≤ 0.1%	
Pump on below 2 dbar	pass	100.0% of scans	≥ 99%	
Sensor calibration age	warn	3 of 9 older than 365 days	≤ 365 days	Temperature 4924: 427 d; PAR (Biospherical/Licor/Chelsea) 70444: 846 d; Fluorometer (WET Labs ECO-AFL/FL) FLRTD 9157: 625 d (1 without a date)
Maximum pressure	info	192.9 dbar		
Downcast descent rate (median)	info	0.20 dbar/s		31.6% of downcast ascending (heave)
Duration	info	27.7 min, 39865 scans		
Bottle fire confirms	info	0		

SP2613_Station1: pressure record and TS diagram





SP2613_Station1: profiles



SP2613_Station2

Sensors

SBE 911plus/917plus CTD; 24 Hz; 0 frequency channels and 0 voltage words suppressed; appended: NMEA position

Channel	Sensor	Serial number	Calibrated	Age at cast [days]
Primary temperature (F0)	Temperature	4924	2025-05-08	427
Primary conductivity (F1)	Conductivity	4537	2026-01-06	184
Pressure (F2)	Pressure (Digiquartz)	0381	2025-09-08	304
Secondary temperature (F3)	Temperature	2059	2026-01-30	160
Secondary conductivity (F4)	Conductivity	4605	2026-02-17	142
Voltage 0	Oxygen (SBE 43)	3357	2025-12-13	208
Voltage 1	PAR (Biospherical/Licor/Chelsea)	70444	2024-03-15	846
Voltage 2	Transmissometer (WET Labs C-Star)	CST 0493DR	2026-02-16	143
Voltage 4	Fluorometer (WET Labs ECO-AFL/FL)	FLRTD 9157	2024-10-22	625
Voltage 6	Altimeter	VA500 88426	—	

Ages over 365 days are highlighted.

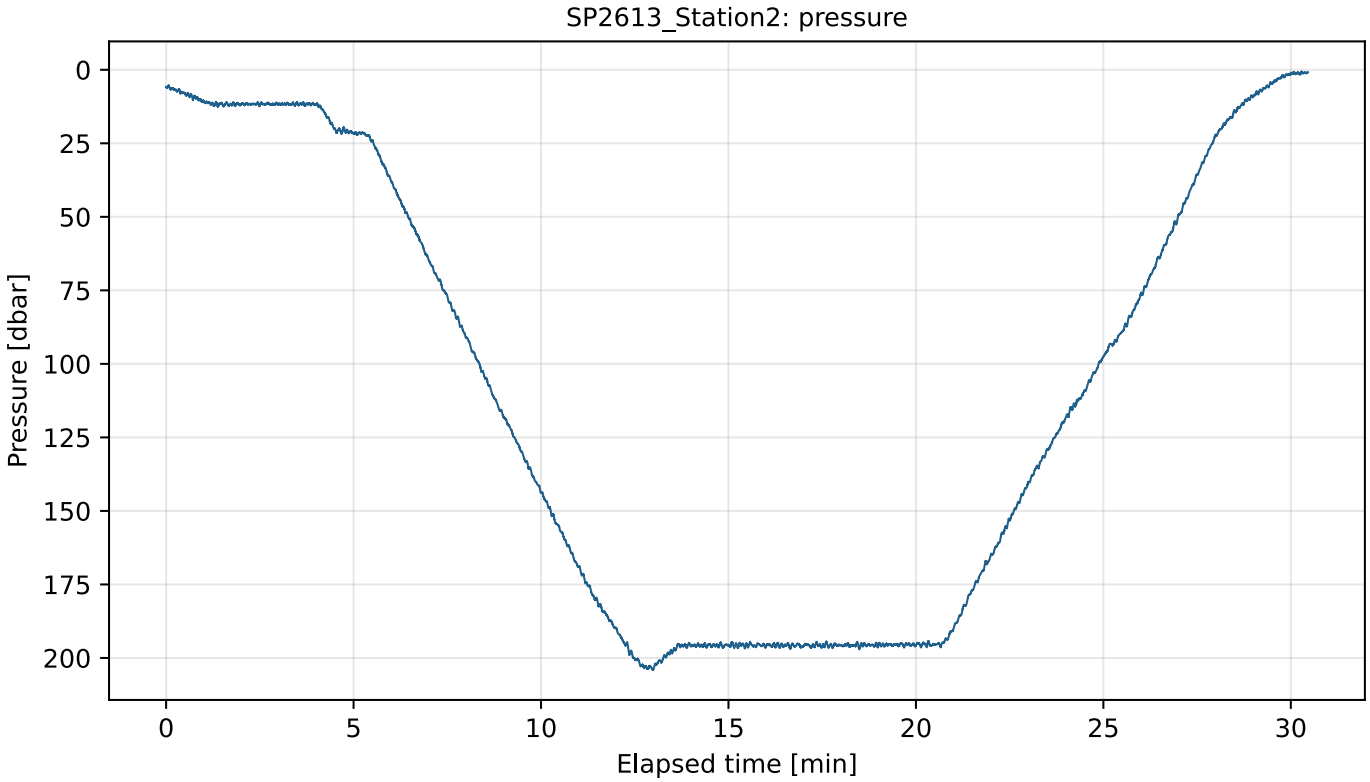
Integrity checks

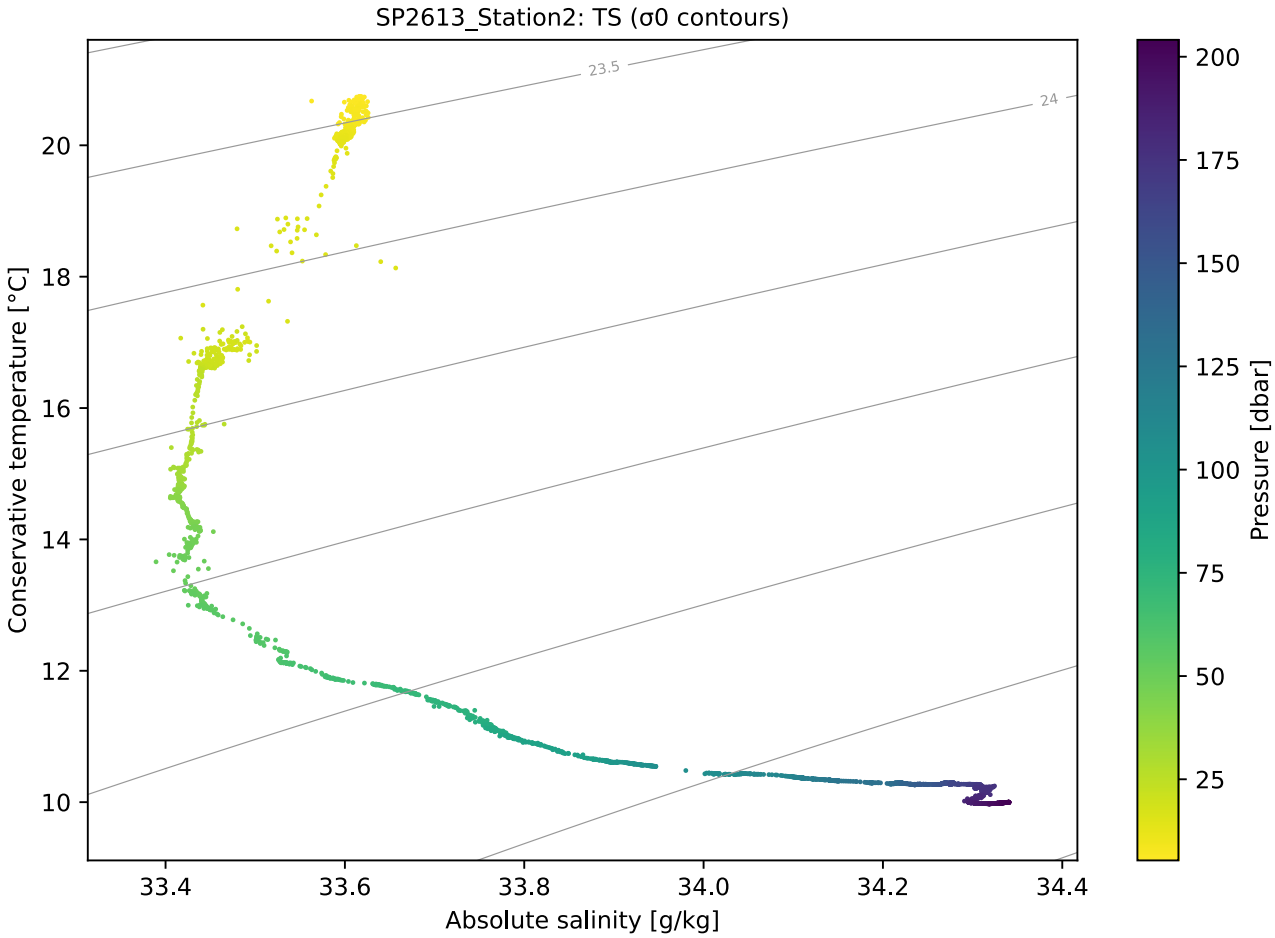
Check	Status	Value	Limit	Detail
Raw files complete	pass	all present		.bl bottle log present
Files readable	pass	43853 scans		
Cast position	pass	32.6010, -117.3655		header
NMEA position on every scan	pass	0 scans without position		
Position inside the expected area	pass	inside	-117.377 to -117.226 E, 32.5997 to 32.7054 N	
Cast time	pass	2026-07-09 19:12:16 UTC		NMEA time
Cast time inside the expected dates	pass	2026-07-09	2026-07-09 to 2026-07-09	
Header serial numbers match the configuration	pass	T 4924, C 4537		

Science checks

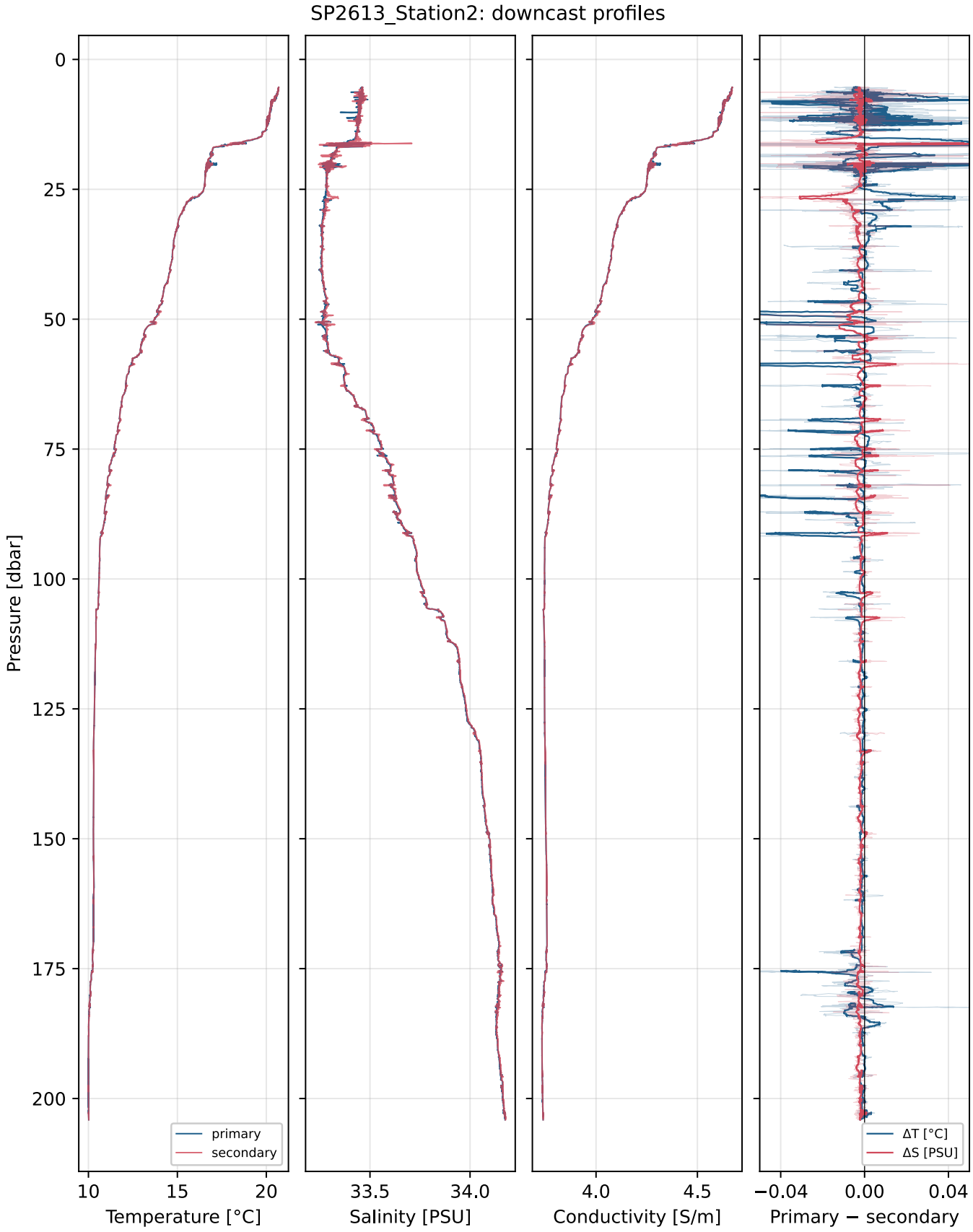
Check	Status	Value	Limit	Detail
Dual temperature difference (median)	pass	0.001252 °C	≤ 0.01 °C	95th percentile 0.05273 °C, below 20 dbar
Dual conductivity difference (median)	pass	0.0003304 S/m	≤ 0.001 S/m	95th percentile 0.004231 S/m, below 20 dbar
Dual salinity difference (median)	pass	0.00265 PSU	≤ 0.01 PSU	95th percentile 0.01296 PSU, below 20 dbar
Pressure range	pass	0.00% outside	-5 to 11000	min 0.5258, max 204.1
Temperature range	pass	0.00% outside	-2.5 to 40	min 9.992, max 20.78
Conductivity range	pass	0.00% outside	0 to 7	min 3.733, max 4.679
Salinity range	pass	0.00% outside	2 to 42	min 33.12, max 34.18
Temperature spikes	pass	0.000% of scans	≤ 0.5%	
Conductivity spikes	pass	0.002% of scans	≤ 0.5%	
Pressure spikes	pass	0.000% of scans	≤ 0.5%	
Lost scans (modulo count)	pass	0 (0.000%)	≤ 0.1%	
Pump on below 2 dbar	pass	100.0% of scans	≥ 99%	
Sensor calibration age	warn	3 of 9 older than 365 days	≤ 365 days	Temperature 4924: 427 d; PAR (Biospherical/Licor/Chelsea) 70444: 846 d; Fluorometer (WET Labs ECO-AFL/FL) FLRTD 9157: 625 d (1 without a date)
Maximum pressure	info	204.1 dbar		
Downcast descent rate (median)	info	0.25 dbar/s		24.5% of downcast ascending (heave)
Duration	info	30.5 min, 43853 scans		
Bottle fire confirms	info	0		

SP2613_Station2: pressure record and TS diagram





SP2613_Station2: profiles



SP2613_Station3

Sensors

SBE 911plus/917plus CTD; 24 Hz; 0 frequency channels and 0 voltage words suppressed; appended: NMEA position

Channel	Sensor	Serial number	Calibrated	Age at cast [days]
Primary temperature (F0)	Temperature	4924	2025-05-08	427
Primary conductivity (F1)	Conductivity	4537	2026-01-06	184
Pressure (F2)	Pressure (Digiquartz)	0381	2025-09-08	304
Secondary temperature (F3)	Temperature	2059	2026-01-30	160
Secondary conductivity (F4)	Conductivity	4605	2026-02-17	142
Voltage 0	Oxygen (SBE 43)	3357	2025-12-13	208
Voltage 1	PAR (Biospherical/Licor/Chelsea)	70444	2024-03-15	846
Voltage 2	Transmissometer (WET Labs C-Star)	CST 0493DR	2026-02-16	143
Voltage 4	Fluorometer (WET Labs ECO-AFL/FL)	FLRTD 9157	2024-10-22	625
Voltage 6	Altimeter	VA500 88426	—	

Ages over 365 days are highlighted.

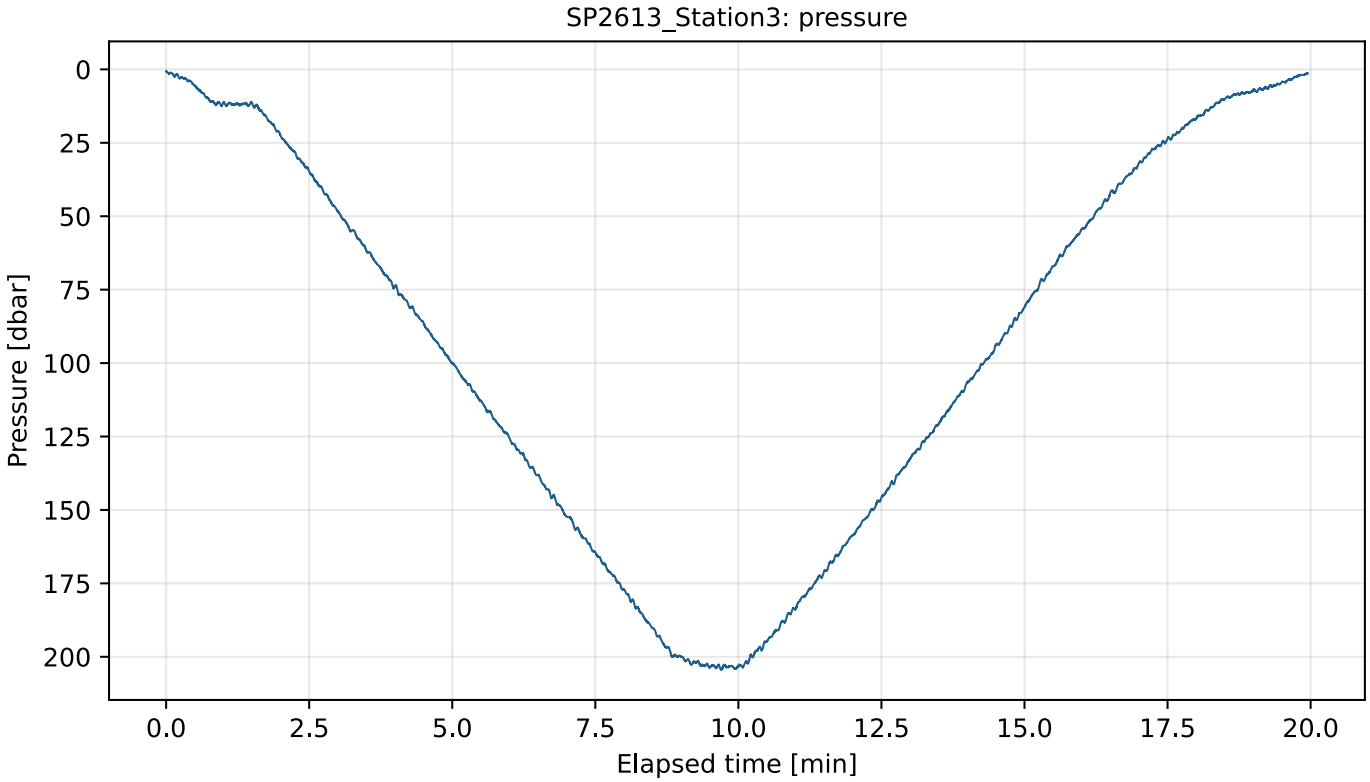
Integrity checks

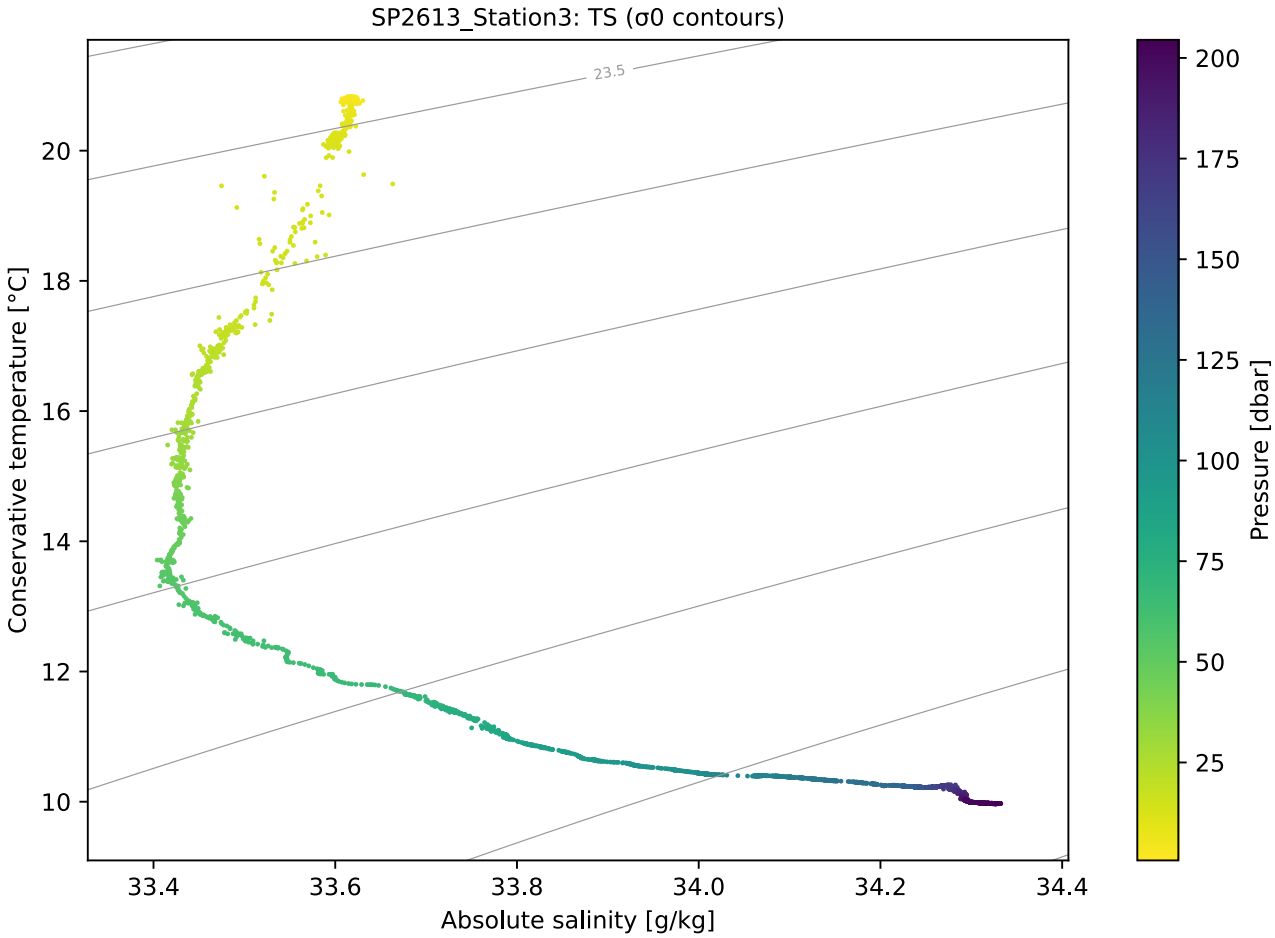
Check	Status	Value	Limit	Detail
Raw files complete	pass	all present		.bl bottle log present
Files readable	pass	28730 scans		
Cast position	pass	32.6022, -117.3742		header
NMEA position on every scan	pass	0 scans without position		
Position inside the expected area	pass	inside	-117.377 to -117.226 E, 32.5997 to 32.7054 N	
Cast time	pass	2026-07-09 20:11:17 UTC		NMEA time
Cast time inside the expected dates	pass	2026-07-09	2026-07-09 to 2026-07-09	
Header serial numbers match the configuration	pass	T 4924, C 4537		

Science checks

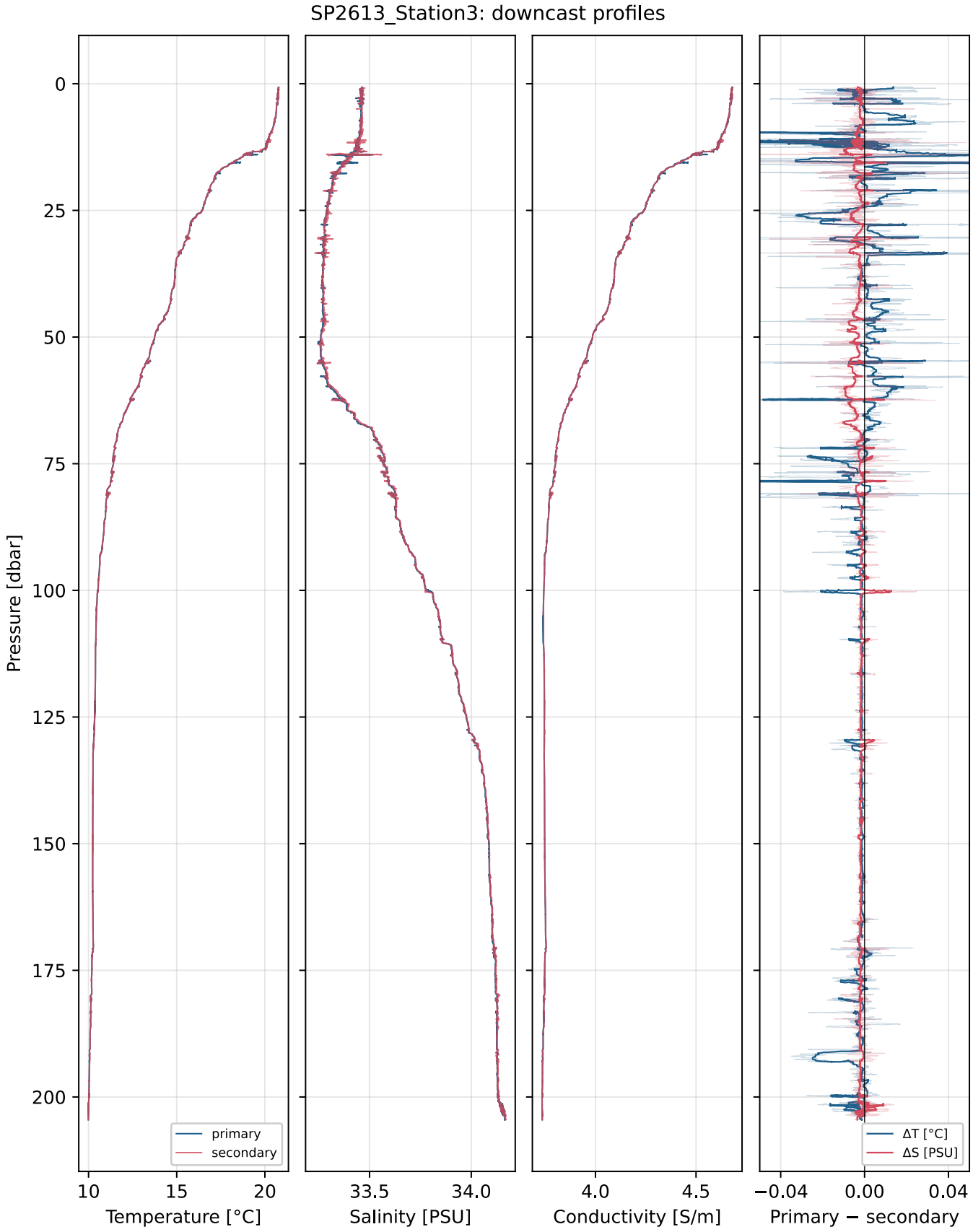
Check	Status	Value	Limit	Detail
Dual temperature difference (median)	pass	0.003251 °C	≤ 0.01 °C	95th percentile 0.06617 °C, below 20 dbar
Dual conductivity difference (median)	pass	0.0003654 S/m	≤ 0.001 S/m	95th percentile 0.005642 S/m, below 20 dbar
Dual salinity difference (median)	pass	0.002602 PSU	≤ 0.01 PSU	95th percentile 0.01537 PSU, below 20 dbar
Pressure range	pass	0.00% outside	-5 to 11000	min 0.6662, max 204.5
Temperature range	pass	0.00% outside	-2.5 to 40	min 9.985, max 20.79
Conductivity range	pass	0.00% outside	0 to 7	min 3.734, max 4.681
Salinity range	pass	0.00% outside	2 to 42	min 33.18, max 34.17
Temperature spikes	pass	0.000% of scans	≤ 0.5%	
Conductivity spikes	pass	0.000% of scans	≤ 0.5%	
Pressure spikes	pass	0.000% of scans	≤ 0.5%	
Lost scans (modulo count)	pass	0 (0.000%)	≤ 0.1%	
Pump on below 2 dbar	pass	100.0% of scans	≥ 99%	
Sensor calibration age	warn	3 of 9 older than 365 days	≤ 365 days	Temperature 4924: 427 d; PAR (Biospherical/Licor/Chelsea) 70444: 846 d; Fluorometer (WET Labs ECO-AFL/FL) FLRTD 9157: 625 d (1 without a date)
Maximum pressure	info	204.5 dbar		
Downcast descent rate (median)	info	0.37 dbar/s		13.1% of downcast ascending (heave)
Duration	info	20.0 min, 28730 scans		
Bottle fire confirms	info	0		

SP2613_Station3: pressure record and TS diagram





SP2613_Station3: profiles



SP2613_Station4

Sensors

SBE 911plus/917plus CTD; 24 Hz; 0 frequency channels and 0 voltage words suppressed; appended: NMEA position

Channel	Sensor	Serial number	Calibrated	Age at cast [days]
Primary temperature (F0)	Temperature	4924	2025-05-08	427
Primary conductivity (F1)	Conductivity	4537	2026-01-06	184
Pressure (F2)	Pressure (Digiquartz)	0381	2025-09-08	304
Secondary temperature (F3)	Temperature	2059	2026-01-30	160
Secondary conductivity (F4)	Conductivity	4605	2026-02-17	142
Voltage 0	Oxygen (SBE 43)	3357	2025-12-13	208
Voltage 1	PAR (Biospherical/Licor/Chelsea)	70444	2024-03-15	846
Voltage 2	Transmissometer (WET Labs C-Star)	CST 0493DR	2026-02-16	143
Voltage 4	Fluorometer (WET Labs ECO-AFL/FL)	FLRTD 9157	2024-10-22	625
Voltage 6	Altimeter	VA500 88426	—	

Ages over 365 days are highlighted.

Integrity checks

Check	Status	Value	Limit	Detail
Raw files complete	pass	all present		.bl bottle log present
Files readable	pass	14187 scans		
Cast position	pass	32.6087, -117.3700		header
NMEA position on every scan	pass	0 scans without position		
Position inside the expected area	pass	inside	-117.377 to -117.226 E, 32.5997 to 32.7054 N	
Cast time	pass	2026-07-09 20:55:05 UTC		NMEA time
Cast time inside the expected dates	pass	2026-07-09	2026-07-09 to 2026-07-09	
Header serial numbers match the configuration	pass	T 4924, C 4537		

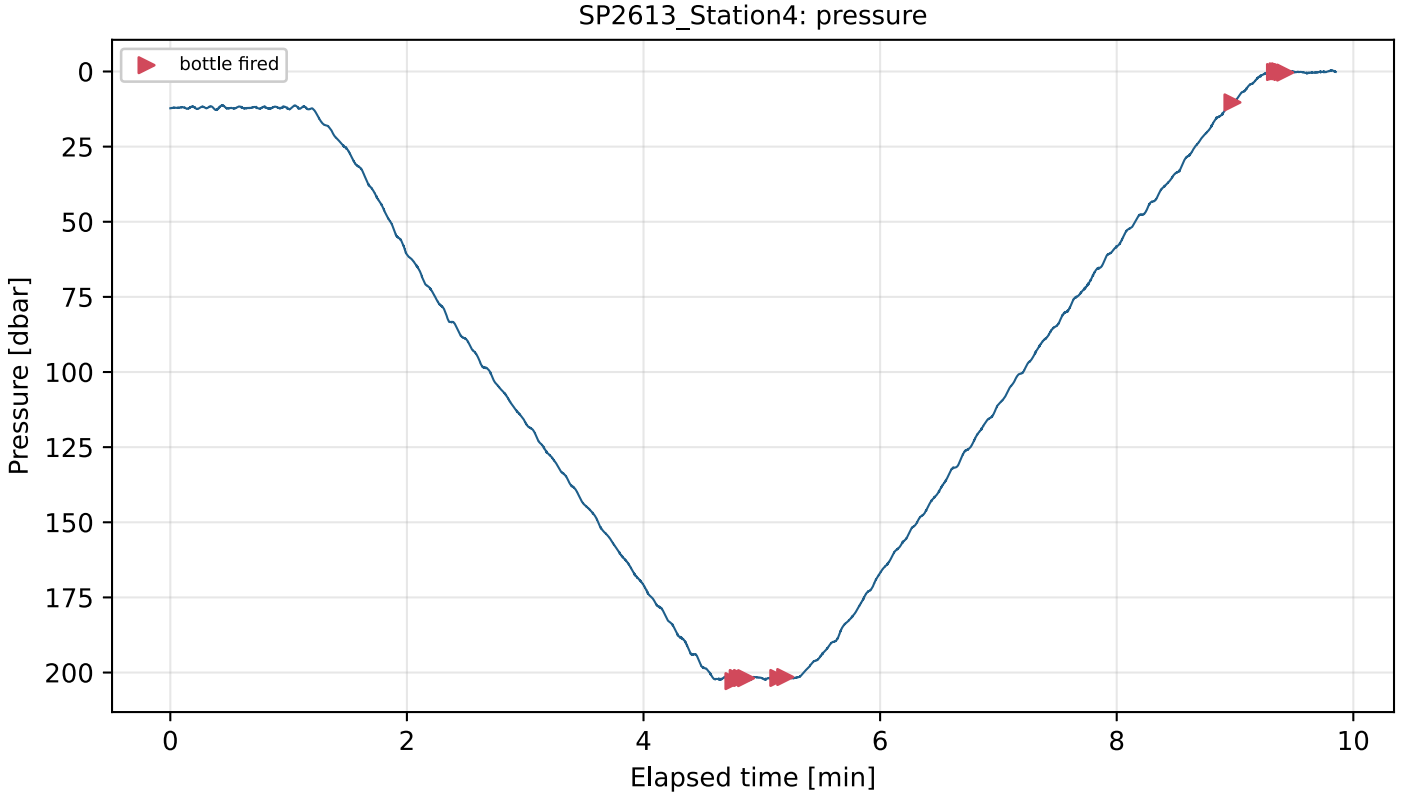
Science checks

Check	Status	Value	Limit	Detail
Dual temperature difference (median)	pass	0.002151 °C	≤ 0.01 °C	95th percentile 0.06393 °C, below 20 dbar
Dual conductivity difference (median)	pass	0.0003301 S/m	≤ 0.001 S/m	95th percentile 0.005409 S/m, below 20 dbar
Dual salinity difference (median)	pass	0.002812 PSU	≤ 0.01 PSU	95th percentile 0.01705 PSU, below 20 dbar
Pressure range	pass	0.00% outside	-5 to 11000	min -0.3758, max 203
Temperature range	pass	0.00% outside	-2.5 to 40	min 9.998, max 20.87
Conductivity range	pass	0.00% outside	0 to 7	min 3.735, max 4.689
Salinity range	pass	0.00% outside	2 to 42	min 33.17, max 34.19
Temperature spikes	pass	0.000% of scans	≤ 0.5%	
Conductivity spikes	pass	0.000% of scans	≤ 0.5%	
Pressure spikes	pass	0.000% of scans	≤ 0.5%	
Lost scans (modulo count)	pass	0 (0.000%)	≤ 0.1%	
Pump on below 2 dbar	pass	100.0% of scans	≥ 99%	
Sensor calibration age	warn	3 of 9 older than 365 days	≤ 365 days	Temperature 4924: 427 d; PAR (Biospherical/Licor/Chelsea) 70444: 846 d; Fluorometer (WET Labs ECO-AFL/FL) FLRTD 9157: 625 d (1 without a date)
Maximum pressure	info	203.0 dbar		
Downcast descent rate (median)	info	0.75 dbar/s		14.8% of downcast ascending (heave)
Duration	info	9.9 min, 14187 scans		
Bottle fire confirms	info	12		

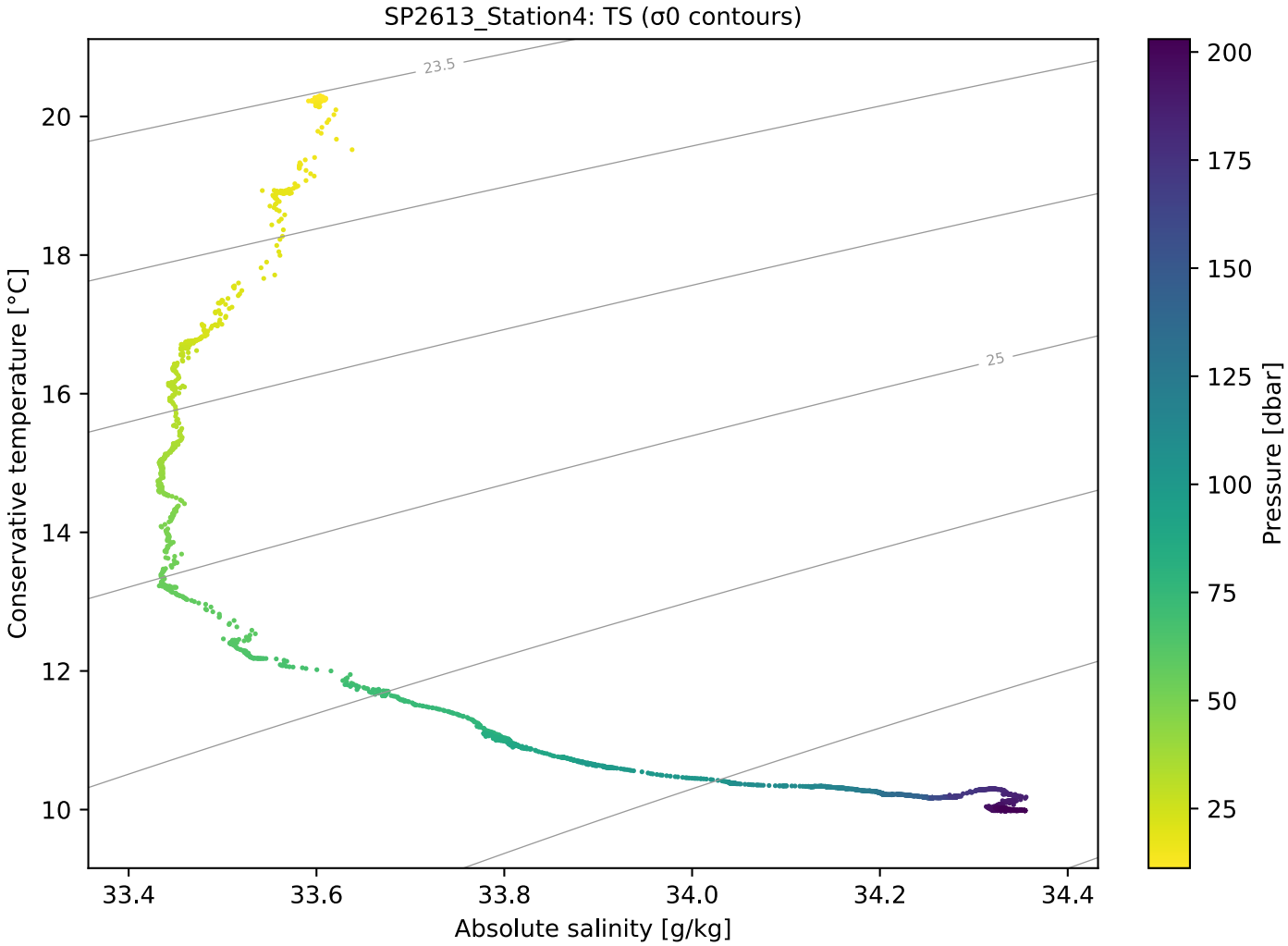
SP2613_Station4: bottle fires and pressure record

Bottle fires (12)

Bottle	Fire order	Time (UTC)	Pressure, mean [dbar]	Range [dbar]
1	1	2026-07-09 20:59:52	202.6	202.2 – 203.0
2	2	2026-07-09 20:59:54	201.9	201.7 – 202.1
3	3	2026-07-09 20:59:56	201.7	201.6 – 201.9
4	4	2026-07-09 20:59:58	202.2	201.8 – 202.5
5	5	2026-07-09 21:00:14	201.9	201.8 – 202.1
6	6	2026-07-09 21:00:18	201.4	201.2 – 201.6
7	7	2026-07-09 21:04:05	10.0	9.7 – 10.3
8	8	2026-07-09 21:04:26	0.1	-0.1 – 0.4
9	9	2026-07-09 21:04:28	0.0	-0.2 – 0.2
10	10	2026-07-09 21:04:29	0.0	-0.2 – 0.4
11	11	2026-07-09 21:04:30	0.3	0.0 – 0.4
12	12	2026-07-09 21:04:31	0.2	0.1 – 0.4



SP2613_Station4: TS diagram



SP2613_Station4: profiles

